Hossam Elsherbiny

AI Engineer

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Professional Summary

AI Engineer with hands-on experience building and deploying machine learning systems for time-series forecasting, NLP, and computer vision. Skilled in Python, TensorFlow, Keras, PyTorch and OpenAI tools, with a strong record of delivering real-time AI applications. Experienced in optimizing models, integrating APIs, and developing end-to-end ML pipelines to solve practical problems with measurable results.

Experience

AI Engineer, Acrow – Cairo — Nov 2024 – Oct 2025

- Extracted and processed data from Oracle databases to support time-series forecasting for over 3,000 SKUs.
- Designed, trained, and evaluated forecasting models (SARIMA, XGBoost, CatBoost, Prophet, LightGBM, and neural networks), lowering forecast error from 80% to 55%.
- Engineered an intuitive GUI-based executable application to automate the end-to-end data extraction, model training, and forecasting pipeline, decreasing the manual process time from 3 days into 1.5 hours.
- Partnered with cross-functional teams to validate model outputs and integrate forecasts into the company's planning processes, boosting profit by over 10.5%.

Education

9-Month Professional Program, Information Technology Institute (ITI) — OCT 2025 – Present

Artificial Intelligence and Machine Learning Track

B.E. in Computer Engineering – Faculty of Engineering, Zagazig University — Oct 2020 – Jul 2025

Ranked 10th in Class.

Projects

EnergiQ Platform | [*GitHub*]

- Deployed an end-to-end, OpenAI-powered AI agent to address user inquiries about electric vehicles, charging stations, and related topics, serving 7,000 users.
- Fulfilled a function-calling framework to interface with the backend database, enabling retrieval of user-specific metrics (e.g., last session details, monthly spending), attained F1-score = 0.94 of Intent Accuracy.
- Enhanced a Retrieval-Augmented Generation (RAG) system to answer general platform, EV, and charging-related queries using internal documentation, achieving Recall@3 = 0.95.
- Enabled voice chat feature using OpenAI APIs (whisper-1, tts-1), reduced inference latency to 78 ms/token.

DeepFM CTR Prediction | [*GitHub*]

- Reproduced the DeepFM paper from scratch in PyTorch on the Criteo dataset ($\sim$45M samples, 39 features), achieving results comparable to published benchmarks (AUC $\approx$ 0.801, LogLoss $\approx$ 0.456).
- Engineered an end-to-end data pipeline (preprocessing, encoding, training, evaluation) handling high-dimensional sparse features, reducing memory overhead via ordinal encoding and efficient embeddings.
- Built and executed 6 ablation experiments (dropout, pooling, batch normalization, skip connections, deep layers, loss functions), identifying performance trade-offs and improving AUC to 0.802 and F1-score to 0.56.
- Analyzed model behavior under extreme class imbalance ($\sim$25% positive class), demonstrating the impact of loss function selection (BCEWithLogits) on recall and classification robustness.

Knowva Platform | [*GitHub*]

- Designed an AI-driven language learning platform that assesses user proficiency through an adaptive exam and personalized content targeting 500k users.
- Integrated OpenAI's Whisper model for real-time speech-to-text conversion, enabling seamless voice input processing and level evaluation with WER = 0.232.
- Optimized the Gemini model to generate personalized responses and custom learning materials, secured a tool response rating of 4.64/5 from users.
- Employed xTTS for text-to-speech functionality and WAV2Lip for lip-sync synchronization, enhancing listening comprehension and pronunciation skills, having 0.92 Cosine Speaker Similarity.

Technical Skills

Programming Languages: Python, SQL, R, C/C++, Java

Analytics Frameworks: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly, Streamlit

ML Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, CatBoost

Computer Vision (CV): OpenCV, YOLO, Mediapipe, Wav2Lip

Natrual Language Processing (NLP): RNNs, LSTMs, GRUs

Generative AI: GPT, Gemini, Llama, AutoEncoder, GANs, Diffusion Models, Hugging Face Transformers

Agentic AI: AI Agents (LangGraph, CrewAI), Advanced RAG (LangChain,Rages), LangSmith

Time Series: ARIMA, ETS, Prophet

MLOps & Tools: Git, Docker, MLflow, Jupyter, Tableau, PowerBI, Spreadsheets, Weights & Biases

Big Data & Cloud: Hadoop, Yarn, Hive, PySpark, AWS(EC2, S3)

Databases: SQLite, MySQL, PostgreSQL, MongoDB, Chromadb, Pgvector

Backend Frameworks: Django, FastAPI

Extracurricular

IEEE Zagazig Student Branch, Computer Society Vice Chairman — OCT 2023 – Jul 2024

- Led a team of 35 novice programmers, instructing them in core algorithms and data structures.
- Mentored participants to successfully complete Level 0 challenges from the ICPC Assiut problem set.
- Contributed to organizing the Mutex event and created problem sets for the Semaphore contest.
- Prepared and coached team members for competitive programming events, including ICPC and IEEEXtreme.

Awards

Second Place, Banha Hackathon (AI Solutions for Language Learning)

Built an AI-driven platform to assess and enhance language proficiency through personalized content and interactive practice.

Third Place, MATE ROV Regional Competition (Co-Pilot & Programming Lead)

Responsible for developing the vision system and implementing image processing algorithms to support underwater task execution.

Additional Information

Military Status **Exempted** (Certificate available upon request).